

The missing Poletsky–Stessin bidisk cyclicity range

An exact later-literature answer

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Status. `literature_already_answered`. This is a literature-identification note, not a claim of a new proof.

Source question

Torkinejad Ziarati’s *Cyclicity in Poletsky–Stessin Weighted Bergman Spaces*, arXiv:2507.03767, Open Problem 1 on page 18, asks:

For $\frac{3}{2} < \beta \leq 2$, determine whether the polynomial

$$f(z_1, z_2) = 1 - \frac{z_1 + z_2}{2}$$

is cyclic in $\mathcal{D}_\beta(\mathbb{D}^2)$.

Here the relevant norm is equivalent to

$$\left\| \sum_{k,l \geq 0} a_{k,l} z_1^k z_2^l \right\|_\beta^2 = \sum_{k,l \geq 0} (k+l+2)^\beta |a_{k,l}|^2.$$

6 Discussion and Open Problems

Comparing our main result on the bidisk with the one in [5], we observe that, in our case, the cyclicity of a polynomial is completely determined by the dimension of its zero set on the distinguished boundary. In contrast, in [5], polynomials of the form $\zeta - z_i$ exhibit a different cyclicity behavior compared to other polynomials that vanish at infinitely many points on $\mathbb{T}^2$.

Open Problem 1. For $\frac{3}{2} < \beta \leq 2$, determine whether the polynomial $f(z) = 1 - \frac{z_1 + z_2}{2}$ is cyclic in $\mathcal{D}_\beta(\mathbb{D}^2)$.

Open Problem 2. Let f be a polynomial that does not vanish on Ω_p and vanishes on a set of real dimension one on $\partial\Omega_p$. Determine under what conditions f is cyclic in $\mathcal{D}_\beta(\Omega_p)$.

Exact supporting answer

Nailwal and Zalar, *Cyclic polynomials in Dirichlet-type Spaces of the unit bidisk*, arXiv:2511.13441, explicitly state in their abstract that they solve this problem. Their Question 1.2 restates it, and Theorem 3.1 proves:

If $\alpha \leq 2$, then $2 - z_1 - z_2$ is cyclic in the sum-weight Dirichlet-type space $\mathcal{D}_\alpha$.

The supporting paper uses weights $(k + l + 1)^\alpha$, while the source uses $(k + l + 2)^\alpha$; the paper explicitly notes that these norms are equivalent. Moreover,

$$1 - \frac{z_1 + z_2}{2} = \frac{1}{2}(2 - z_1 - z_2),$$

and multiplication by a nonzero scalar does not change the closed shift-invariant subspace generated by a polynomial. Thus Theorem 3.1 gives a full affirmative answer to the exact source problem, including the endpoint $\beta = 2$.

Theorem 3.1. *Let $\alpha \leq 2$. Then $p(z_1, z_2) = 2 - z_1 - z_2$ is cyclic in $\mathcal{D}_\alpha$.*

Proof. It suffices to show that p is cyclic in $\mathcal{D}_2$. Let $f \in \mathcal{D}_2$ be such that $f \perp [p]$. Consider the following series of f ,

$$f(z_1, z_2) = \sum_{i,j=0}^{\infty} \frac{b_{i,j}}{(i+j+1)^2} z_1^i z_2^j.$$

We will show that $f = 0$, or equivalently $b_{i,j} = 0$ for $i, j \in \mathbb{Z}_+$. By $f \perp [p]$, it follows that

$$2b_{k,l} = b_{k+1,l} + b_{k,l+1}, \quad k, l \in \mathbb{Z}_+. \quad (3.1)$$

Their Theorem 1.3 is stronger: for $1 < \alpha \leq 2$, an irreducible polynomial without zeros in the bidisk is cyclic precisely when its zero set on $\mathbb{T}^2$ is empty or finite.

Search and scope

The local run indexes and bounded web searches used arXiv:2507.03767, the exact polynomial, the paper title, **Poletsky--Stessin**, and the missing parameter range. The later answer was found through a 2026 cyclic-polynomial reference and verified directly in arXiv:2511.13441. The supporting authors explicitly know they are solving the source problem.

This identification answers Open Problem 1 only. The source's Open Problem 2 about one-dimensional boundary zero sets on complex ellipsoids is not covered.

References

1. P. Torkinejad Ziarati, *Cyclicity in Poletsky--Stessin Weighted Bergman Spaces*, arXiv:2507.03767 (2025).

2. R. Nailwal and A. Zalar, *Cyclic polynomials in Dirichlet-type Spaces of the unit bidisk*, arXiv:2511.13441 (2025; revised 2026).